



# Traffic Signal Optimization Pilot Proposal

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**Location: Shenzhen, China**

**Website: <https://enertraffic.com>**

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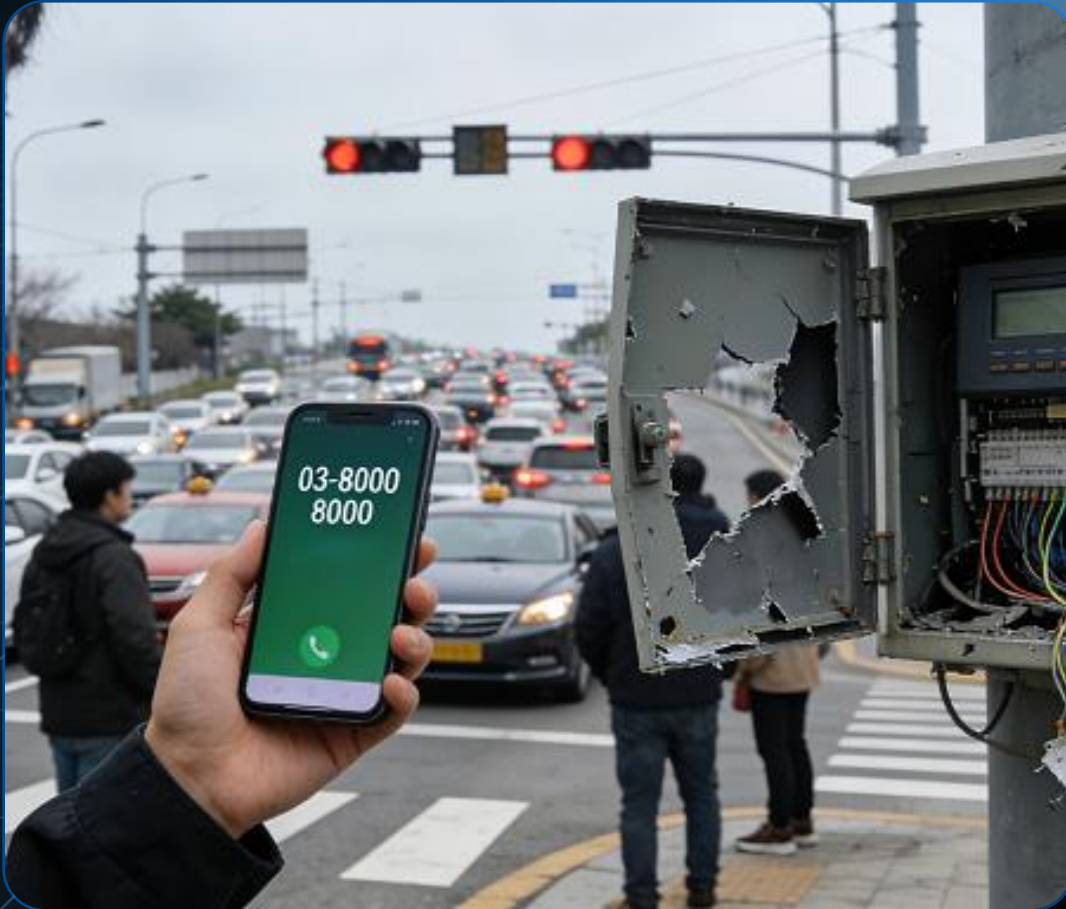


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01

# Challenges of Traffic Management

# Challenges in Traffic Signal Management



## Unscientific Timing

Relies on experience without data models, leading to resource waste. Fixed timing fails to adapt to dynamic traffic conditions.



## Limited Sensing

Traditional sensors offer only simple recognition and limited functionalities. High maintenance costs and long recovery times hinder comprehensive monitoring (e.g. loop traffic detector).



## Passive Management

Lack of proactive monitoring means problems are only addressed after complaints, preventing timely improvements.



## Isolated Intersections

Lack of coordination between intersections prevents optimal traffic flow distribution.

02

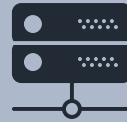
# Traffic Light Optimization Pilot Solution

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# Intelligent Transportation Architecture Overview

## Unified Management Layer

Signal Control Platform · Data Fusion · Solution Diagnosis Regional Coordination · App Enablement



Unified Signal Control Platform

## Precision Control Layer

Built-in Green Wave Coordination & Single-Point Optimization · Real-Time Response · Safety & Stability



Intelligent Signal Controller Edge Computing Hub

## Holistic Perception Layer

All-Weather · Full-Element · Flow · Speed · Queue · Trajectory · Event · Vehicle Type



Camera



Radar



Fusion Sensor



RSU



Env. Sensor



Beidou&GPS

# Traffic Light Optimiztion Pilot Soution

**Solution Principle:** Simple and secure, Isolated from the existing network, Easy to interconnect with the existing network, Smooth Scaling and evolution.



Cloud Platform



## Data Sensing Layer

Deploy radar-video fusion sensors to collect real-time traffic data to provide accurate information for .precise control of traffic lights.



## Precision Control Layer

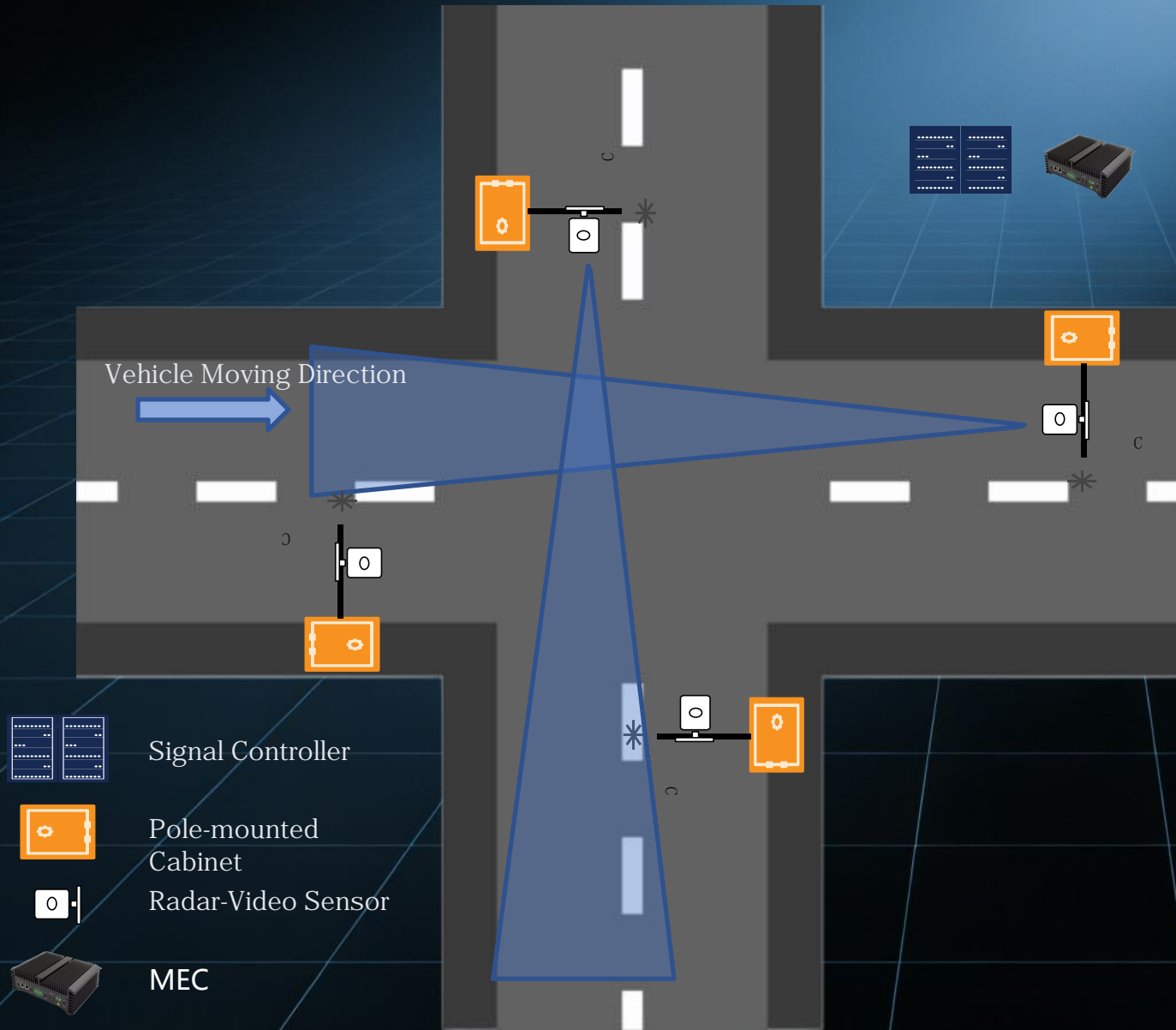
Signal controller + MEC execute optimization strategies, achieving a closed-loop adaptive system from perception to control.



## Unified Mng. Layer

Provides unified traffic management system and interconnection with surrounding systems.

# Deployment Diagram



| Device Type               | Qty          |
|---------------------------|--------------|
| Signal controller         | 1            |
| Radar-video fusion sensor | 4            |
| Switch                    | 4            |
| Pole-mounted cabinet      | 4            |
| MEC                       | 1            |
| Fill light                | 4 (Optional) |

1. A signal controller is required for the entire intersection to control all traffic signal lights at the intersection.
2. 4 radar-video fusion devices are to be installed to face traffic flow at the intersection.
3. An MEC may be configured and installed inside the signal controller cabinet for edge computing.

# Core Equipment: Signal Controller



## Technical Specifications

- 32 light groups, 96 output terminals & 32 detector inputs
- 32-channel leakage detection & IP65 protection rating
- 12-inch touch screen for easy operation
- Double doors design with manual control & backup power

## Key Functions

- Adaptive control
- Multi-period control
- Networked coordination
- Supports manual operation
- Special control modes

# Key Advantages of Signal Controller



## High Security

Stable operation with low failure rate. Equipped with multiple safety mechanisms including leakage protection, independent yellow flash, configurable degradation strategies etc.



## High Compatibility

Supports standard protocols for seamless integration with upper-level platforms. Support 3rd sensors access. Opens private protocols to execute third-party optimization algorithms.



## Abundant Functions

Supports rich functions including daily operations, intelligent optimization, and regional coordination.

# Core Equipment: Smart Radar



## ⚙️ Technical Specifications

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- 📶 Working frequency: 80GHz
- 📏 Detection Range: 350m+
- 🕒 Detectable targets: 200+
- 📷 Resolution: 5MP for traffic detection and LPR, Starlight night vision

## 🏆 Key Advantages

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- 🔄 Radar + video fusion technology for enhanced accuracy
- 🌐 Comprehensive Traffic Data Acquisition
- 🕒 Easy Implementation & Maintenance

# Core Equipment: MEC

## Key Value Proposition

Enabling real-time local data processing and collaborative decision-making at the edge. This ensures low-latency responses and efficient handling of critical traffic data right where it's generated.



## Technical Specifications

-  **Processor**  
4×Cortex-A76 + 4×Cortex-A55, 2.4GHz
-  **System Memory**  
32G DDR5
-  **Operating System**  
Linux
-  **Input Voltage**  
220V
-  **Interfaces**  
USB3.0/HDMI/Ethernet/RS485

# Key Advantages of MEC



## Flexible Computing Power

With computing power scalable up to 200 TOPS, the MEC can be flexibly configured to meet the demands of simple to highly complex traffic scenarios.



## Built-in Core Algorithms

Pre-installed with mature algorithms for video cross-domain fusion, green wave coordination, and single-point optimization etc, significantly shortening deployment cycles.



## Rich Interfaces & Expandability

Equipped with USB, HDMI, Ethernet, RS485, and 4G/5G modules, ensuring broad compatibility and easy future expansion.

# Single Intersection Optimization Procedure

## Single-point Adaptive Optimization

Creates a closed-loop control system from front-end perception, adaptive control, to system evaluation. This improves efficiency and solves issues like uneven green time distribution and wasted green time.

## Optimization Process



Front-end Perception

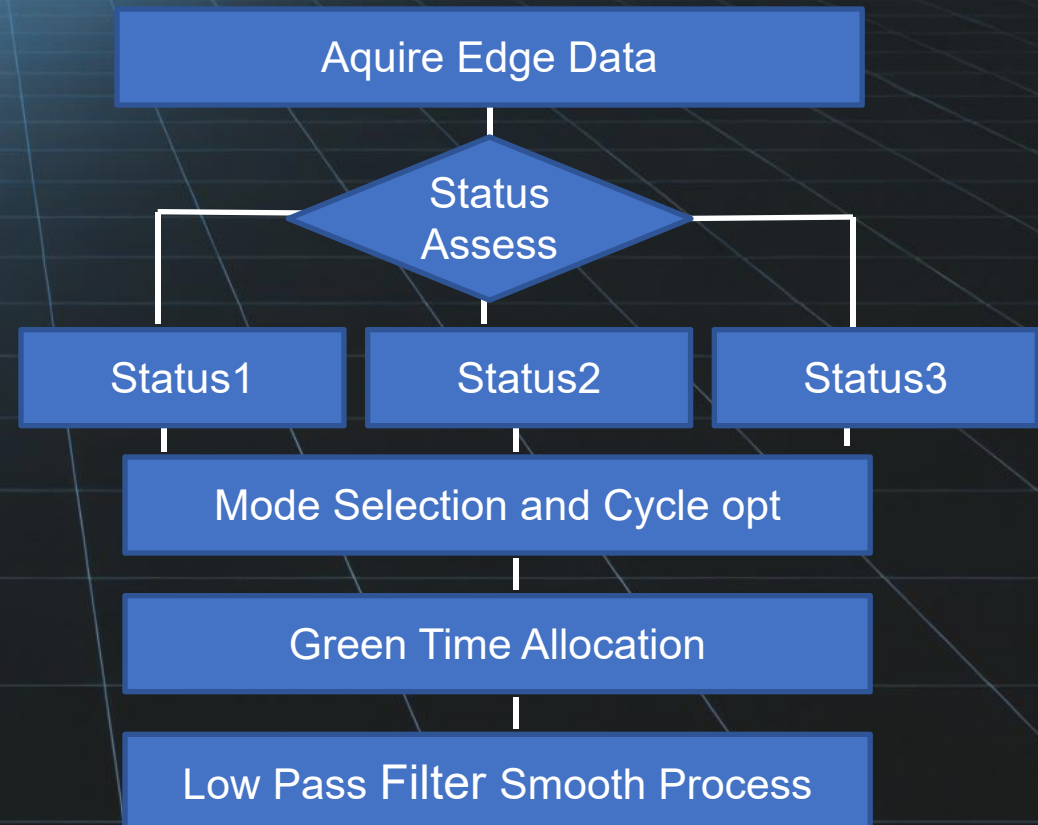


Adaptive Signal Control



Traffic Evaluation & Refinement

## System Flowchart



# Front-end Perception



## Temporal Statistical Data

- Traffic flow monitoring
- Time occupancy analysis
- Vehicle headway measurement



## Spatial Statistical Data

- Queue length detection
- Zone vehicle count
- Spatial occupancy rate



**Core Capability:** Accurately detects road traffic flow information by acquiring multi-dimensional temporal and spatial data.

# Signal Control



## Timing Prediction

Acquires temporal statistics to predict optimal timing plans, laying the foundation for efficient traffic flow.



## Real-time Control

Uses real-time spatial data for precise phase switching, responding instantly to current traffic conditions.



## Adaptive Adjustment

Automatically adjusts phase split online based on detector data, enabling fully adaptive control strategies.

# Traffic Evaluation & Refinement

## Evaluation Indicators



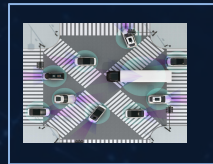
## Core Capability

The system evaluates optimization effectiveness by comparing traffic parameters before and after implementation. This comprehensive analysis ensures that we can quantify the impact of our strategies and drive continuous improvement in traffic management.

## Multi-Dimensional Effectiveness Evaluation



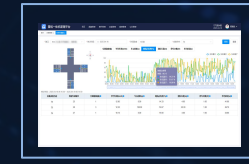
Safety Dimension



Efficiency Dimension



Equity Dimension



System Performance  
Dimension

# Multi-source Data Fusion

- Phase, Stage, and Cycle Data Collection
- Automatic Timing Plan Validation

Signal Controller Integration

## Timing Plan Collection

Captures phase, stage, and cycle data from signal controller timing plans

## Timing Plan Validation

Auto-validates plan accuracy using real-time traffic flow data

- All-Weather Capability (Rain/Snow/Night)
- Comprehensive Perception of People, Vehicles, and Roads

Radar-Video Fusion



- Vehicle Detection: Lane-level precise positioning with multi-type classification
- Vehicle Tracking: Trajectory tracking for accurate flow calculation in complex lanes
- Vehicle Classification: Distinguishes cars, vans, buses, trucks, and more

- Comprehensive: Full-Space Modeling.
- Granular: Lane-Level and Signal-Cycle-Level Metrics
- Accurate: 95%+ Accuracy for Traffic Flow and Key Metrics

Indicator Calculation

|  | #  | Indicator Name              |
|--|----|-----------------------------|
|  | 1  | Traffic Volume              |
|  | 2  | Average vehicle speed       |
|  | 3  | Congestion Evaluation       |
|  | 4  | Queue length                |
|  | 5  | Time headway                |
|  | 6  | Space headway               |
|  | 7  | Space occupancy rate        |
|  | 8  | Time occupancy rate         |
|  | 9  | Queue Dissipation Time      |
|  | 10 | Saturation Flow Rate        |
|  | 11 | Lost Time                   |
|  | 12 | Queue Length at Green Onset |
|  | 13 | Saturation Degree           |

# Typical Intelligent Traffic Light Scenarios



## Inductive Control Mode

For small to medium urban intersections, detects vehicle presence to adjust signal timing efficiently.



## Adaptive Control Mode

For large urban intersections, uses real-time radar-video data to generate dynamic timing plans, solving queuing issues.



## Dynamic Green Wave Mode

For urban arterial roads, maximizes green wave bandwidth based on real-time data to enhance corridor efficiency.

# E2E Safety Assurance

## Hardware Security



- **Hard Yellow Flash:** Dedicated board for critical exceptions.
- **Surge Protection:** Robust defense against lightning/strikes.
- **Hardware Redundancy:** Backup critical circuit boards.
- **Backup Power:** Continuous operation during outages.
- **Leakage Protection:** Built-in hazard detection.

## Software & Logic Security



- **Soft Yellow Flash:** Software-triggered on logical errors.
- **Config Self-Check:** Automatic integrity verification.
- **Proactive Prevention:** Logic safeguards against errors.
- **Safe Startup:** Secure initialization to avoid unsafe states.

## System-Level Security



- **Environmental Monitoring:** Tracks temp/humidity for reliability.
- **Degradation & Fault-Tolerant Operation:** Auto-detection of module failures (detectors, coordination, etc.), Smooth degradation to safer preset modes (E.G, Coordinated → Single-point adaptive etc), Prevents intersection disruption.

# Timing Plan Management



## Plan Access & Modification

Automatic Access and Update of Timing Plans  
Access of Full Scheduling Plans and Time-of-Day  
Tables



## Diagnosis & Timing Recommendations

Our system analyzes plans and provides intelligent  
recommendations for optimal timing adjustments.



## Manual Secondary Editing

Fine-tune timing plans with a user-friendly interface  
for manual adjustments and customization.



## Historical Plan Query

Effortlessly retrieve and compare historical timing  
plans to track changes and performance over time.

03

## Pilot Effect Evaluation

# Pilot Project Effect Evaluation

## Indicator Overview and Analysis

Core KPIs (Volume, Speed, Delay Index) with MoM/WoW comparison — intuitively showing intersection performance trends.

## Key Indicator Ranking Analysis

Ranked by Volume, Queue Length, and Avg Delay — Quickly pinpoint congestion hotspots.

## Space Resource Comparison

Saturation vs. Road Space (Lanes, Width) — Evaluate efficiency, identify mismatches.

## Trend and Pattern Analysis

Charts for daily/weekly/monthly trends — Cyclical patterns & anomaly detection

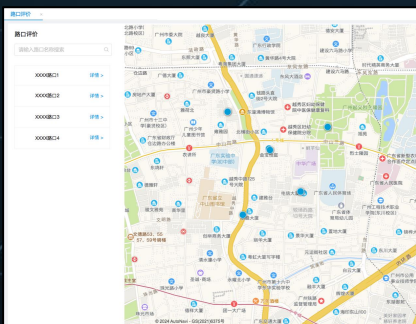
## Time-Period Characteristic Evaluation

- Peak, off-peak, low-peak analysis
- Pinpoint bottleneck periods
- Inform differentiated signal timing

## One-Click Export of Multi-Dimensional Data

- Export in multiple formats
- Support analysis, reporting, archiving

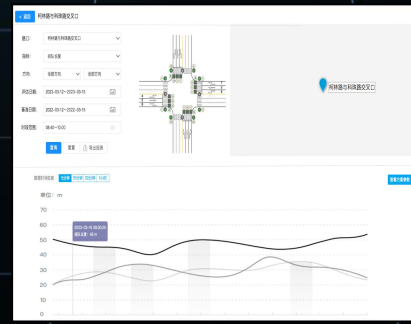
(PoP) Indicator Analysis



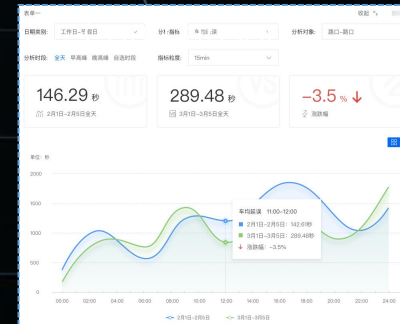
Space Resource Comparison



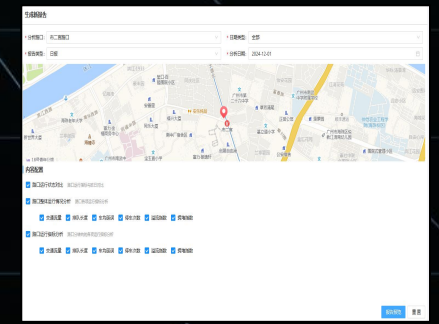
Trend Comparison



Time-Period



Evaluation Report



# 04

## Implementation and Deployment Plan

# Implementation and Deployment Plan

## Phase 1 (W1-2): Preparation

Site Survey, Plan Confirmation,  
Equipment Preparation

## Phase 2 (W3): Installation

Installation, cabling, and  
power-up of all hardware  
devices

## Phase 3 (W4-5): Commissioning

Radar calibration, lane  
mapping, and system  
integration testing.

## Phase 4 (W6-8): Data Collection

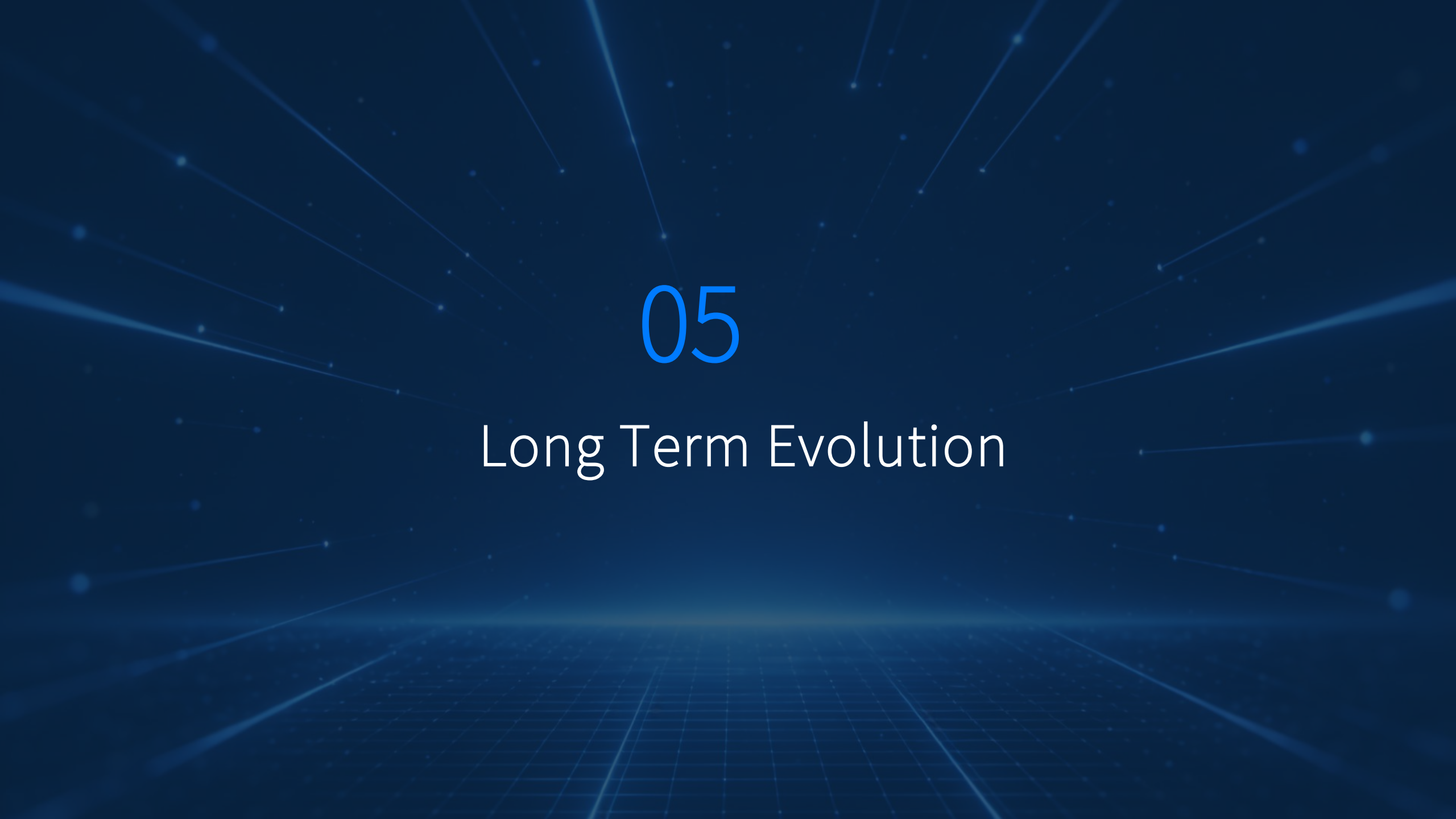
Operate in traditional fixed-  
time mode to collect baseline  
data for comparison.

## Phase 5 (W9-10): Validation

Switch to adaptive mode to  
collect optimized data.

## Phase 6 (W11): Conclusion

Conduct comprehensive data  
comparison and analysis, and  
output report.

The background is a deep blue gradient. A perspective grid of thin, light blue lines extends from the bottom towards the center, creating a sense of depth. Numerous thin, light blue lines radiate outwards from the center towards the edges of the frame. Small, faint blue dots are scattered throughout the background, particularly along the radiating lines.

05

# Long Term Evolution

# Future Evolution Path



## Intersection Enhancement

Introduce cutting-edge technologies such as Vehicle-to-Everything (V2X) to enhance single-point perception and control capabilities.



## Intersection Coordination

Single-point to multi-intersection coordinated control enabling regional-level traffic coordination and management.



## Integrated O&M

Integrate with existing signal controller networks to achieve unified device access and efficient operations and maintenance management.



## Cross-Department Collaboration

Break down data silos to enable data sharing and interact with public security, special services, and other departments.

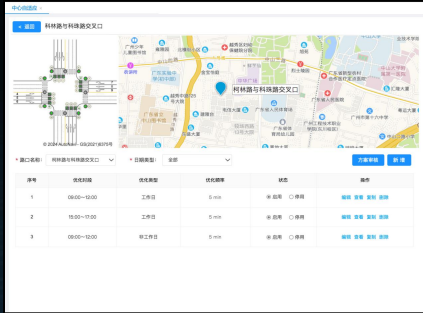


## Towards a City Brain

Build a globally optimal intelligent traffic decision-making center based on full-scale data and AI algorithms.

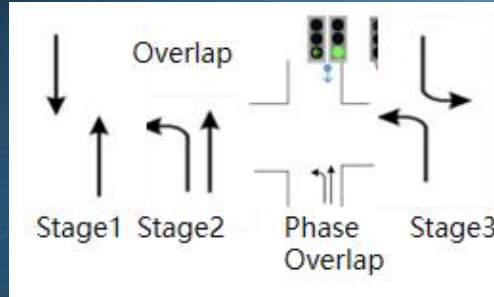
# Single → Multi-Intersection Coordination EnerTraffic

## Single intersection enhancement



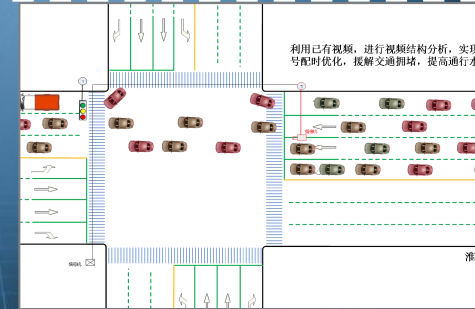
### Single-Point Enhancement

Fine-tune single-point timing plans based on real-time multi-dimensional information.



### Phase Overlap

Flexibly adjust phase overlap while maintaining the phase sequence within the ring.



### Overflow Prevention & Control

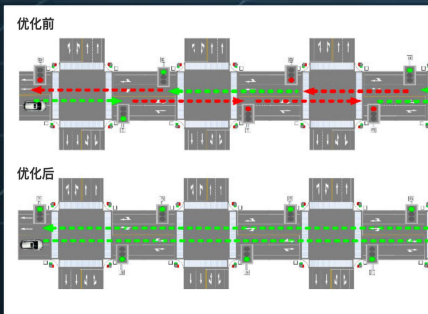
Monitor overflow risks in real time and implement relief measures at bottleneck intersections to prevent traffic gridlock.



### Variable Lane Control

Adjust variable lane directions in real time to improve intersection efficiency from a spatial perspective.

## Multi-Intersection Coordination&Optimization



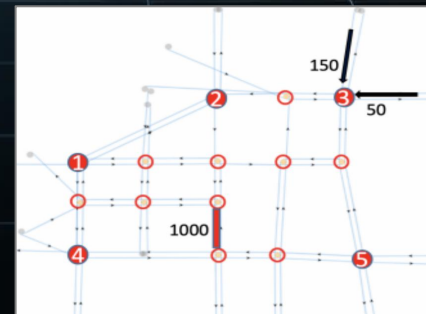
### Arterial Coordination Control

Adjust intersection offsets to enable green wave passage and improve arterial traffic efficiency.



### Green Wave Optimization

Speed-based offset fine-tuning with periodic split adjustment for smooth, stable progression.



### Arterial Metering Control

Upstream control during peak hours to protect oversaturated areas and reduce central city congestion.

## Regional Traffic Optimization



### Regional Traffic Demand Mng.

Boundary control nodes regulate entry based on real-time congestion during peak hours.

# Unified Signal Control

Integrated Signal  
Network Control

Four Unifications

Data Fusion &  
Applications

## Unified Business Mng.

- Unified interface for all operations
- Cross-brand special service coordination
- Cross-brand coordinated control

## Unified Signal Mng.

- Interconnection via standard interfaces for unified management.

## Unified Interface Mng.

- Integrate with existing traffic systems
- Enable Vehicle-Road-Cloud integrated applications

## Unified Data Services

- Multi-source data fusion
- Standardized data sharing mechanism

 Signal Timing and Status Push

 Integration with Superior Platform

 Indicator Data Collection

 Multi-Source Data Fusion

 Signal Optimization Effectiveness Evaluation

 Multi-Dimensional Statistical Analysis

 Networking & Integrated Control

 Point-Line-Area Coordinated Optimization

 Public Transport Priority

 Special Service Assurance

 Multi-Brand Signal Controller Compatibility

 Signal Status and Plan Monitoring

 Intersection Inventory

 Signal Controller O&M

# Cross-Department Collaboration Toward the City Brain



## Special Event Security

- Real-time data + intelligent signal control
- Dynamic route planning for VIPs & major events
- Ensure safe and smooth passage



## Emergency Management

- Rapid emergency vehicle identification
- Signal priority control
- Open a "green corridor" for critical response



## Municipal Urban Management

- Traffic impact assessment for road works
- Route optimization for sanitation vehicles
- Nighttime construction noise monitoring & control



## Traffic Enforcement

Utilizes high-definition video and AI recognition technology to automatically capture various traffic violations (such as running red lights, illegal parking, etc.), improving enforcement efficiency and fairness.



# THANKS

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Should you need further information or have any questions about this proposal, please feel free to contact us.

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